

LumbiniX Hackathon Project Report

"Path of Wisdom" – Complete Step-by-Step Implementation Guide

Document Information

Field	Details
Project Name	Path of Wisdom
Event	National LumbiniX Hackathon 2026
Theme	"Where Spirituality Meets Innovation"
Question	"What If Buddha Were Born in 2026?"
Duration	48 Hours
Platform	Mobile App (React Native + Expo)

1. Project Overview

"Path of Wisdom" is an AR-powered pilgrimage app where users:

1. Walk around Lumbini (Pokémon Go-style map)
 2. Receive **Buddhist riddles** randomly or at map locations
 3. **Guess the location** from the riddle and walk there
 4. Tap "**I Found It**" – if wrong, get a **Buddha-style motivational hint**
 5. If correct → **AR camera opens** with floating items
 6. Tap items → **Collect TOOL + WISDOM** → Stored in **Inventory**
 7. **Side quests** (meditate, photo, clean, write) unlock more wisdom
 8. **Wisdom Levels** unlock via main + side quests
 9. **Buddha Chat** – Socratic questioning (never direct answers)
 10. **Jarvis-style AR overlay** – point camera at heritage sites → info appears
 11. **Upload work** (cleaning, etc.) → AI detects → **Blockchain token** rewarded
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2. Complete Feature List

#	Feature	Description
1	Interactive Map (Pokémon Go-style)	GPS-based map showing heritage sites
2	Riddle Discovery	Riddles appear randomly OR at map locations

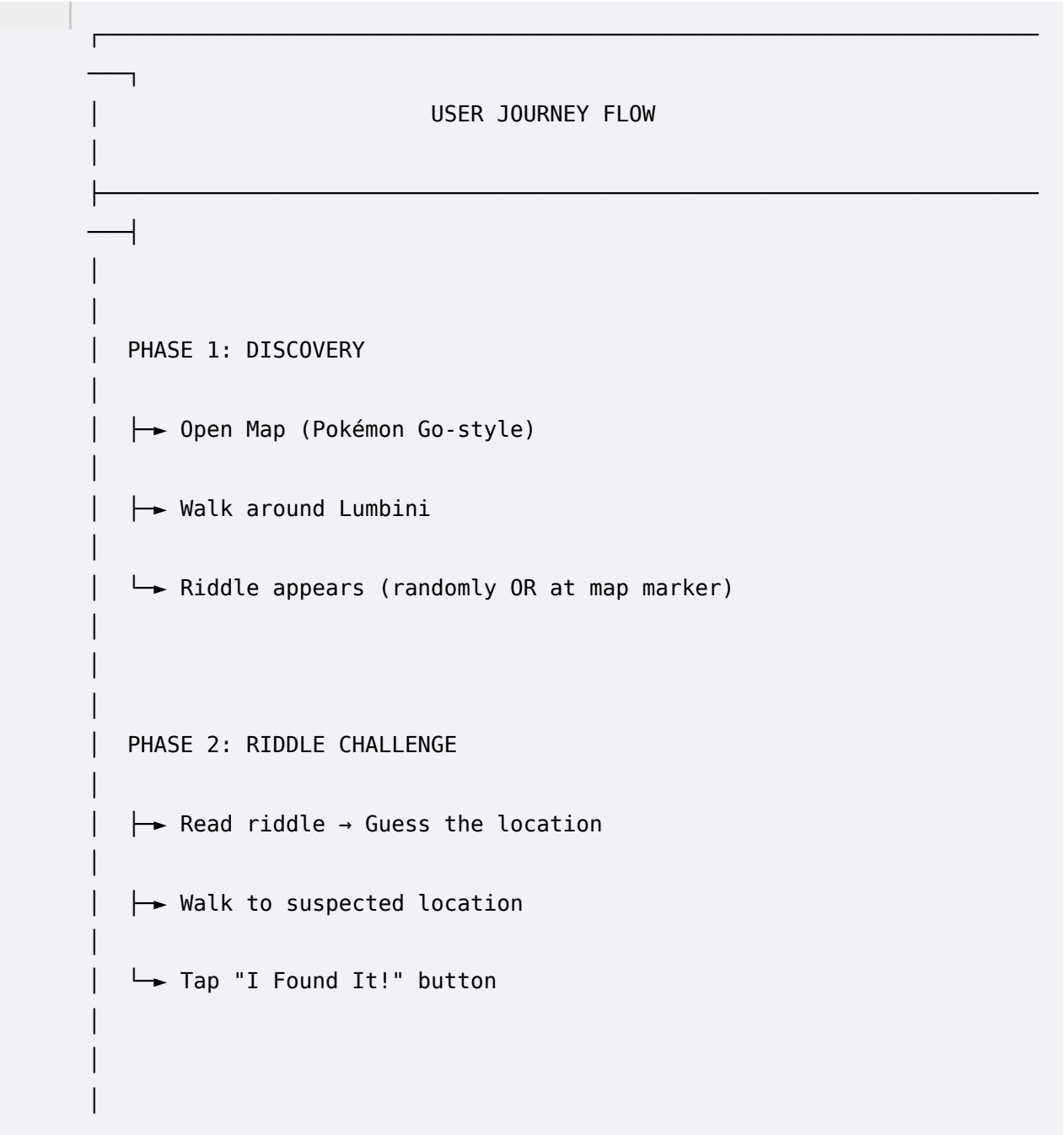
3	"I Found It" Verification	User taps → GPS checks if at correct location
4	Buddha Hint System	Wrong location → motivational hint in Buddha's style
5	AR Item Collection	Virtual items appear in AR camera at correct location
6	Wisdom Unlock	Each item collected unlocks Buddha's wisdom teaching
7	Inventory System	Both items AND wisdom stored

		permanently
8	Side Quests	Meditate, photo, clean, write tasks → unlock wisdom
9	Wisdom Levels	3 levels: Foundation → Deepening → Awakening
10	Jarvis-Style AR Heritage Guide	Camera overlay with site info when pointed at heritage
11	Buddha Wisdom Chat	AI Buddha with Socratic questioning (no direct answers)

12	Upload & Token Rewards	Upload work → AI detects → Blockchain token rewarded
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3. Complete User Journey

text



PHASE 3: VERIFICATION

- └─> App checks GPS location
- └─> ❌ WRONG: "Seek further, traveler..." → Hint appears
- └─> ✅ CORRECT: AR camera opens

PHASE 4: AR COLLECTION

- └─> Floating items appear in AR view
- └─> Tap on items to collect
- └─> TOOL added to inventory ✅
- └─> WISDOM popup appears with Buddha teaching 🌸

PHASE 5: INVENTORY STORAGE

- └─> ITEM saved (name, type, icon, date, location)
- └─> WISDOM saved (text, source, level, date)
- └─> View anytime in Inventory Screen

PHASE 6: JARVIS-STYLE HERITAGE GUIDE (Always Available)

- └─> Point camera at any heritage site
- └─> GPS detects nearby site
- └─> Information overlay appears on screen (Jarvis-style)

PHASE 7: SIDE QUESTS

↳ Tasks: Meditate, Photo, Clean, Write

↳ Complete → More wisdom unlocked

↳ Progress → Higher wisdom levels

PHASE 8: BUDDHA CHAT

↳ Tap Buddha avatar → Ask any question

↳ AI responds with Socratic questions (never direct answers)

PHASE 9: UPLOAD & TOKENS

↳ Upload photo of good work (cleaning, etc.)

↳ AI detects and verifies

↳ Blockchain token rewarded 

4. Riddle Examples

Riddle	Answer	Hint (if wrong)
"Under this tree, a	Mayadevi Temple	"Look for a garden

prince took seven steps and declared, 'I am the chief of the world.' Where am I?"		with a sacred pond and ancient sal trees..."
"The emperor who gave up war and carved his teachings on stone. My pillars stand tall. Where am I?"	Ashoka Pillar	"Look for a stone pillar with a lion on top, near the temple..."
"From Japan, a white stupa rises for world peace. Where am I?"	World Peace Pagoda	"Look for a white dome- shaped structure with golden spire..."

"The sacred pool where the queen bathed before giving birth. Where am I?"	Pushkarini Pond	"Look for a rectangular pond near the temple..."
"The tree where the Buddha attained enlightenment in a past life. Where am I?"	Bodhi Tree	"Look for a large tree with heart-shaped leaves..."

5. What Gets Stored in Inventory

Items Collected:

text

```
{
  id: "meditation_bell",
  name: "Meditation Bell",
  type: "tool",
  icon: "🔔",
  collectedAt: "2026-08-08T10:30:00Z",
  siteId: "mayadevi_temple"
}
```

Wisdom Unlocked:

text

```
{
  id: "wisdom_1",
  text: "Mindfulness begins with a single breath...",
  source: "Mayadevi Temple",
  level: 1,
  collectedAt: "2026-08-08T10:30:00Z"
}
```

6. Wisdom Level System

Level	Name	Teaching Focus	Requirements
1	Foundati on	Four Noble Truths, Noble Eightfold Path	Visit 3 sites, collect 3 items
2	Deepenin g	Dependen t Origination, Five Aggregate s	Visit 5 sites, collect 5 items
3	Awakenin g	Emptiness (Sunyata), Buddha Nature	Visit 8 sites, collect 8 items

7. Complete Technology Stack

Component	Technology	Open-Source Library	License	Link
App Framework	React Native + Expo	Expo CLI	MIT	expo.dev
AR Engine	ViroReact	@reactvision/react-viro	MIT	npmjs.com/package/@reactvision/react-viro
AR Starter Kit	Expo + TypeScript	expo-starter-kit-typescript	MIT	viro-community.readme.io
Geospatial Anchors	ViroReact + ReactVision	rvApiKey + rvProjectId	Platform Feature	viro-community.readme.io/docs/geospatial
Map & GPS	react-native-maps + expo-location	react-native-maps	MIT	github.com/react-native-maps/react-native-maps

Local Storage	AsyncStorage	@react-native-async-storage/async-storage	MIT	-
AI Chat	Karma Electric-8B	anicka/karma-electric-llama31-8b	Apache-2.0	huggingface.co/anicka/karma-electric-llama31-8b
AI Serving	Ollama	ollama	MIT	ollama.ai
Blockchain	Ethereum/Arbitrum	ethers.js, RainbowKit	MIT	docs.ethers.io
3D Models	GLB/GLTF	Poly Haven, Sketchfab (free)	CC0	polyhaven.com
AI Detection	Google Cloud Vision / TensorFlow.js	google-cloud-vision	Apache-2.0	cloud.google.com/vision

8. Step-by-Step Implementation Plan

Phase 1: Project Setup (Hours 1-3)

Step	Task	Tool/Library	What to Do
1	Create Expo project	Expo CLI	<code>npx create-expo-app PathOfWisdom --template</code>
2	Install ViroReact	@reactvision/react-viro	<code>npm install @reactvision/react-viro</code>
3	Clone ViroReact Starter Kit	expo-starter-kit-typescript	<code>git clone https://github.com/ReactVision/expo-starter-kit-typescript</code>
4	Install Map + Location	react-native-maps, expo-location	<code>npm install react-native-maps + npx expo</code>

			install expo- locatio n
5	Install Storage	AsyncStor age	npm install @react- native- async- storage/ async- storage
6	Configure app.json	Expo Config Plugin	Add ["@r eactvisi on/react - viro"] t o plugins array
7	Get ReactVisi on Credential s	ReactVisi on Studio	Sign up at studio.r eactvision .xyz → Get rvApiKey + rvProjectI d
8	Prebuild	Expo	npx expo

			prebuild d
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Critical Note: ViroReact works with both React Native CLI and Expo projects. For Expo projects, you must use a development build or `expo prebuild` .

Phase 2: Interactive Map & GPS (Hours 4-10)

Step	Task	Tool/Library	What to Do
1	List heritage sites	Data file	Create <code>heritageSites.js</code> with Lumbini site coordinates
2	Build map component	react-native-maps	Display map with site markers
3	Add GPS tracking	expo-location	<code>Location.watchPositionAsync()</code> for continuous tracking

4	Proximity detection	Custom logic	Check if user within 50m of site → trigger riddle
5	Riddle appears	UI	Show riddle card on map when near location or randomly

Reference Project: Similar AR archaeology apps use GPS and magnetometer to guide users through heritage sites and trigger pop-up information windows .

Lumbini Heritage Sites:

- Mayadevi Temple (27.4699°N, 83.2757°E)
- Ashoka Pillar (27.4695°N, 83.2761°E)
- World Peace Pagoda (27.4725°N, 83.2790°E)

Phase 3: AR Quest & Item Collection (Hours 11-22)

Step	Task	Tool/Library	What to Do
1	Set up AR scene	ViroReact	Create AR QuestScene.tsx with ViroARScene

2	Add 3D items	GLB/GLTF	Place models in <code>assets/models/</code>
3	Floating animation	ViroAnimations	Register <code>float</code> animation for items
4	Tap-to-collect	Viro3DObject	<code>onClick</code> triggers collection + wisdom
5	Wrong location hint	Custom logic	Show Buddha-style motivational hint
6	Wisdom popup	React Native	Display Buddha teaching on collection
7	Save to inventory	AsyncStorage	Store <code>{items: [], wisdom: []}</code>

ViroReact Features:

- AR plane detection with horizontal and vertical surface anchors

- Geospatial Anchors that pin content to real-world latitude, longitude, and altitude
- `hostGeospatialAnchor()` → Persist anchors to backend for sharing
- `resolveGeospatialAnchor()` → Resolve anchors hosted by others

Jarvis-Style AR Overlay Implementation:

Step	Task	What to Do
1	Host Geospatial Anchors	Use <code>hostGeospatialAnchor(lat, lng, alt, "Terrain")</code> for each heritage site
2	Resolve Anchors	Use <code>resolveGeospatialAnchor(platformUuid)</code> when user opens camera near site
3	Display Info Cards	Use <code>ViroText + V</code>

		<code>iroImage</code> anchored to site location
4	GPS Proximity	Use <code>rvFindNearbyGeospatialAnchors(lat, lng, radius, limit)</code> to find nearby sites

ViroReact Expo Starter provides 7 pre-built scenes including geospatial anchors, 3D object loading, and plane detection .

Phase 4: Riddle & Quest Discovery (Hours 23-30)

Step	Task	Tool/Library	What to Do
1	Create riddle database	JSON	Store riddles + answers + location hints
2	Random walking trigger	GPS	Riddle appears when near

			any site OR randomly
3	"I Found It" button	UI	User taps → GPS verification
4	Hint system	Custom logic	Wrong location → show hint (Buddha style)
5	Side quests	Data file	Meditate, photo, clean, write tasks

Phase 5: Inventory & Wisdom Levels (Hours 31-37)

Step	Task	Tool/Library	What to Do
1	Inventory data structure	JavaScript	Define <code>items[]</code> and <code>wisdom[]</code> objects

2	Save function	AsyncStorage	<code>addToInventory(item, wisdom)</code>
3	Inventory screen	React Native	Display items and wisdom with tabs
4	Level logic	JavaScript	Track progress: 3 sites → Level 1, 5 → Level 2, 8 → Level 3
5	Level badges	React Native	Show current level on inventory

Phase 6: AI Buddha Chat (Hours 38-43)

Step	Task	Tool/Library	What to Do
1	Install Karma Electric	Ollama	<code>ollama pull anicka/karma-electric</code>

			llama31-8b
2	Create Modelfile	Ollama	Set system prompt: "You NEVER give direct answers..."
3	Run model	Ollama	ollama run karma-electric-v3
4	Build chat UI	React Native	Show Buddha avatar + chat bubbles
5	Connect to backend	Fetch API	Send user messages → get AI response

Karma Electric-8B Details:

- Fine-tuned Llama 3.1 8B model
- 4,092 training examples (v9, February 2026)
- 87.03% accuracy
- Apache 2.0 license (free)

- Includes Bodhisattva Axis activation capping for persona stabilization
- Recommended: Use soft capping with `alpha=0.5` to preserve nuance

System Prompt (Socratic):

"You are the voice of Lord Buddha. You NEVER give direct answers. You ALWAYS respond with reflective questions that guide the user to discover wisdom themselves."

Phase 7: Upload, AI Detection & Blockchain (Hours 44-47)

Step	Task	Tool/Library	What to Do
1	Image upload	expo-image-picker	User takes photo of work (cleaning, etc.)
2	AI detection	Google Cloud Vision / TensorFlow.js	Analyze image → detect if work is valid
3	Token reward	ethers.js + RainbowKit	Mint token on quest completion
4	Smart contract	Solidity	ERC-1155 soul-bound token (non-

			transferable)
5	IPFS storage	ipfs-http-client	Store token metadata

9. Open-Source Library Quick Reference

Library	License	Installation	Purpose
ViroReact	MIT	<code>npm install @reactvision/react-viro</code>	AR features
Expo Starter Kit	MIT	<code>git clone https://github.com/ReactVision/expo-starter-kit-typescript</code>	Pre-built AR scenes
Karma Electric-8B	Apache-2.0	<code>ollama pull anicka/karma-</code>	AI Buddha chat

		electric - llama31- 8b	
react-native-maps	MIT	npm install react- native- maps	Map display
expo-location	MIT	npx expo install expo- locatio n	GPS tracking
AsyncStorage	MIT	npm install @react- native- async- storage/ async- storage	Inventory storage
ethers.js	MIT	npm install ethers	Blockchai n interaction
Rainbow Kit	MIT	npm install @rainbow -	Wallet UI

		me/rainbowkit	
Poly Haven	CC0	polyhaven.com	Free 3D models

10. 48-Hour Timeline Summary

Phase	Hours	Tasks	Key Libraries
Setup	1-3	Project create, ViroReact, dependencies	Expo CLI, ViroReact
Map & GPS	4-10	Heritage sites, map, GPS, riddles	react-native-maps, expo-location
AR Collection	11-22	AR scene, 3D items, inventory	ViroReact, AsyncStorage
Quest Discovery	23-30	Riddles, "I Found It", hints, side quests	-

Inventory & Levels	31-37	Inventory screen, level logic	AsyncStorage
AI Chat	38-43	Karma Electric, Socratic prompt	Ollama, Karma Electric
Upload & Tokens	44-47	Image upload, AI detection, blockchain	ethers.js, RainbowKit
Polish	48	Testing, demo video, slides	-

11. Critical Path (Must Have for Demo)

1.  Map with GPS and heritage markers
 2.  Riddle appears (random or location-based)
 3.  "I Found It" verification (GPS check)
 4.  Wrong location → Buddha hint
 5.  Correct location → AR camera opens with floating items
 6.  Tap to collect → ITEM + WISDOM saved
 7.  Inventory shows items and wisdom
 8.  Jarvis-style AR overlay at heritage sites
 9.  Side quests (at least 2 working)
 10.  Wisdom levels (basic progression)
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12. Risk Mitigation

Risk	Mitigation
AR not working on device	Use ViroReact Expo Starter (pre-tested)
AI model too slow	Use quantized GGUF (Q8_0, 8.5GB) with Ollama
GPS inaccurate	Use expo-location with high accuracy
3D models too large	Use GLB compressed from Poly Haven
Time running out	Focus on critical path, defer full blockchain

Note: ViroReact does NOT support Expo Go. You MUST use a development build .

13. Demo Day Checklist

- ☐
Map shows with GPS working
 - ☐
Riddle appears → User walks → "I Found It" works
 - ☐
Wrong location → Hint appears
 - ☐
Correct location → AR opens with items
 - ☐
Tap item → Wisdom popup appears
 - ☐
Inventory shows both items and wisdom
 - ☐
Jarvis-style overlay works when camera pointed at heritage site
 - ☐
Side quest completes → wisdom unlocks
 - ☐
Buddha chat responds (Socratic)
 - ☐
Prepare 2-3 minute demo video
-

14. References & Resources

Resource	Link	Purpose
ViroReact Docs	viro-community.readme.io	AR implementation

ViroReact Geospatial Anchors	viro-community.readme.io/docs/geospatial	Pin AR to GPS
ViroReact Expo Starter	github.com/ReactVision/expo-starter-kit-typescript	Pre-built AR scenes
ViroReact NPM Package	npmjs.com/package/@reactvision/react-viro	Installatio n
Karma Electric	huggingface.co/anicka/karma-electric-llama31-8b	AI chat
Ollama	ollama.ai	Running AI locally
Poly Haven	polyhaven.com	Free 3D models
ReactVisi on Studio	studio.reactvision.xyz	Get API keys

Tracking & Anchors Guide	viro-community.readme.io/docs/tracking-and-anchors	AR plane detection
ReactVision Studio Guide	updates.reactvision.xyz	Scene setup tutorial

15. Summary

Aspect	Details
Discovery	Riddles (random/location) → Guess location → Walk → "I Found It"
Verification	GPS check → Wrong? → Buddha hint → Keep searching

AR Collection	Correct? → Camera opens → Items float → Tap to collect
AR Heritage Guide	Point camera at any site → Jarvis- style info overlay
Storage	ITEM + WISDOM saved to inventory permanen tly
Progressi on	Main quests + Side quests → Wisdom levels unlock
Chat	Buddha avatar → Socratic questionin g (no

	direct answers)
Tokens	Upload work → AI detects → Blockchain token rewarded

All tools are open-source and free. Nothing needs to be built from scratch.

Report prepared for LumbiniX Hackathon 2026

Team "Path of Wisdom"

 **Good luck with your hackathon!**

This response is AI-generated, for reference only.